

ABSTRACT BOOK



13TH INTERNATIONAL Coral Reef Symposium

19–24 JUNE 2016 • HONOLULU, HAWAII

This program is produced for information and to provide assistance on site at the meeting. It contains the most up-to-date scientific program information through 25 May 2016. All changes received after the printing of the program will be updated on the conference website and will be included in the meeting mobile app. Additional information also can be found on the conference website.

WWW.SGMEET.COM/ICRS2016



Satyanarayana, C., Zoological Survey of India, IIInd Floor, Fire Proof Spirit Building, Indian Museum Complex, Kolkata-16, India, chowdula_narayana@rediffmail.com

Chandran, R., Coral Transplantation/restoration project, Zoological Survey of India, CCF Office campus, Jamnagar-1, India, softcoralchandran@gmail.com

Chandra, K., Zoological Survey of India, M-Block, New Alipore, Kolkata-53, India, kailash611@rediffmail.com

Tikadar, S., Gulf of Kachchh Marine National Park, Jamnagar-1, India, stikadar@gmail.com

Yogesh Kumar, J. S., Project Young Scientist, Zoological Survey of India, Port Blair, India, coralyo-geeth@gmail.com

<mailto:stikadar@gmail.com>

RESTORATION OF EXTIRPATED SCLERACTINIAN CORALS IN TURBID ENVIRONMENT AT GULF OF KACHCHH, INDIA – A SUCCESS STORY (Abstract ID: 28656)

Gulf of Kachchh, the northern most limit of coral distribution in the Indian Ocean, inhabits temperature tolerant and most adopted residual coral species. Up-liftment of tectonic plates of the Gulf and transgression of sea levels made all branching corals to extirpate. Semi-arid conditions, large semi-diurnal tidal amplitude, coral exploitation in the past, high sedimentation, industrialization and anthropogenic activities degraded the reefs and hampering the resilience and recovery capacity of the reefs. During 2013-2015, a total 1000 sq. m. area was restored in 3 spatially distinct sites, along the Southern Gulf using jeopardized native coral species and branching coral species brought from Gulf of Mannar, about 2600 km. far. Site specific innovative and modified restoration methods and modern technologies including remote-sensing were used in each and every step of reef rehabilitation. Maximum survival rate of native coral transplants has been recorded in Pirotan (97.24%) and minimum in Narara (96.50%). Likewise maximum survival rate of branching coral species has been observed in Mithapur (85.25%) and was least in Pirotan (68.75%). Among the native transplants, *Cyphastrea serailia* recorded highest mean extension rate followed by *Porites* lichen (5.81 ± 2.12 cm/yr and 5.42 ± 1.25 cm/yr respectively). Among branching coral transplants, *Acropora formosa* and *A. intermedia* recorded highest mean extension rates (8.04 ± 1.17 cm/yr and 7.13 ± 1.84 cm/yr respectively). Damming reduced sedimentation and site specific strategies are the prime reasons for restoring 10,000 years back extirpated staghorn corals successfully in the Gulf.

Sauafea-Le'au, F., NOAA Fisheries PIRO, American Samoa, fatima.sauafea-leau@noaa.gov